

Easy (10)

1.

What layer do VLANs operate on?

- A) Layer 1
- B) Layer 2
- C) Layer 3
- D) Layer 4

Answer: B

Explanation: VLANs operate at the Data Link layer (Layer 2).

2.

Which protocol is primarily used for VLAN tagging?

- A) ISL
- B) 802.1Q
- C) STP
- D) RIP

Answer: B

Explanation: IEEE 802.1Q is the standard VLAN tagging protocol.

3.

What is the default VLAN on most Cisco switches?

- A) 0
- B) 1
- C) 100
- D) 4094

Answer: B

Explanation: VLAN 1 is the default VLAN.

4.

What does a VLAN do?

- A) Creates a separate IP subnet
- B) Creates a separate broadcast domain
- C) Routes packets between switches
- D) Assigns IP addresses

Answer: B

Explanation: VLANs create separate broadcast domains.

5.

What type of port carries traffic for only one VLAN?

- A) Trunk port
- B) Access port
- C) Native port
- D) Routed port

Answer: B

Explanation: Access ports belong to a single VLAN.

6.

What VLAN ID range is valid in 802.1Q?

- A) 1-100
- B) 1-4094
- C) 0-4095
- D) 1-1024

Answer: B

Explanation: VLAN IDs range from 1 to 4094.

7.

What does VTP stand for?

- A) VLAN Trunking Protocol

- B) Virtual Trunking Protocol
- C) VLAN Transfer Protocol
- D) Virtual Transport Protocol

Answer: A

Explanation: VTP stands for VLAN Trunking Protocol.

8.

What device is needed to route traffic between VLANs?

- A) Switch
- B) Router
- C) Hub
- D) Repeater

Answer: B

Explanation: Routers or Layer 3 switches perform inter-VLAN routing.

9.

Native VLAN traffic is:

- A) Tagged
- B) Untagged
- C) Dropped
- D) Encrypted

Answer: B

Explanation: Native VLAN traffic is sent untagged.

10.

Which VLAN ID is reserved and cannot be used for user VLANs?

- A) 0 and 4095
- B) 1 and 100
- C) 100 and 200
- D) 4094 and 4095

Answer: A

Explanation: VLAN IDs 0 and 4095 are reserved.

Medium (30)

11.

Which VLAN type can a port carry traffic from multiple VLANs?

- A) Access
- B) Trunk
- C) Routed
- D) Management

Answer: B

Explanation: Trunk ports carry multiple VLANs.

12.

What VLAN mode allows a switch port to automatically negotiate trunking?

- A) Access
- B) Dynamic Auto
- C) Dynamic Desirable
- D) Both B and C

Answer: D

Explanation: Both dynamic auto and dynamic desirable negotiate trunking.

13.

What is the default VTP mode on Cisco switches?

- A) Server
- B) Client
- C) Transparent
- D) Off

Answer: A

Explanation: Default is Server mode.

14.

Which VLAN membership type assigns VLANs based on MAC address?

- A) Port-based
- B) Protocol-based
- C) MAC-based
- D) Dynamic

Answer: C

Explanation: MAC-based VLAN assigns based on device MAC.

15.

What command shows VLAN information on Cisco switches?

- A) show vlan brief
- B) show interfaces vlan
- C) show ip route
- D) show spanning-tree

Answer: A

Explanation: show vlan brief lists VLANs.

16.

How many VLANs can 802.1Q support?

- A) 128
- B) 256
- C) 1024

D) 4094

Answer: D

Explanation: 802.1Q supports up to 4094 VLANs.

17.

Which of the following is a benefit of VLANs?

- A) Larger collision domains
- B) Reduced broadcast traffic
- C) Increased IP addresses
- D) Physical separation required

Answer: B

Explanation: VLANs reduce broadcast traffic by segmentation.

18.

Which protocol prevents loops in VLAN networks?

- A) VTP
- B) STP
- C) DTP
- D) RIP

Answer: B

Explanation: Spanning Tree Protocol prevents loops.

19.

Which command assigns a port to VLAN 10?

- A) switchport mode trunk
- B) switchport access vlan 10
- C) vlan 10
- D) interface vlan 10

Answer: B

Explanation: Assigns port to VLAN 10.

20.

What happens if two switches have different native VLANs configured on the same trunk?

- A) Trunk link fails
- B) Traffic on native VLAN is dropped or misrouted
- C) VLANs merge
- D) No effect

Answer: B

Explanation: Native VLAN mismatch causes issues.

21.

What is VLAN pruning?

- A) Blocking unused VLANs on trunks
- B) Removing VLANs from switch
- C) Combining VLANs
- D) Changing VLAN IDs

Answer: A

Explanation: VLAN pruning restricts VLAN traffic on trunk links.

22.

What does "VLAN hopping" refer to?

- A) Moving devices between VLANs physically
- B) Attacker accessing VLANs via tagging tricks
- C) Upgrading VLANs
- D) Increasing VLAN numbers

Answer: B

Explanation: VLAN hopping is a security attack.

23.

Which VLAN ID is commonly used as the native VLAN?

- A) 0
- B) 1
- C) 100
- D) 4094

Answer: B

Explanation: Default native VLAN is 1.

24.

What is the main purpose of DTP?

- A) Manage VLAN databases
- B) Dynamically negotiate trunk links
- C) Block VLAN hopping
- D) Route VLAN traffic

Answer: B

Explanation: DTP negotiates trunking automatically.

25.

How is inter-VLAN routing commonly implemented?

- A) Using a router with subinterfaces
- B) Using VLAN pruning
- C) Using multiple switches
- D) Using broadcast domains

Answer: A

Explanation: Router-on-a-stick uses subinterfaces.

26.

Which of these VLAN types isolate ports within the same VLAN?

- A) Private VLANs
- B) Management VLANs
- C) Voice VLANs

D) Data VLANs

Answer: A

Explanation: Private VLANs isolate ports.

27.

What does the VTP domain define?

A) A set of switches sharing VLAN info

B) VLAN ID range

C) Trunk port type

D) Access port mode

Answer: A

Explanation: VTP domain shares VLAN config.

28.

What is the maximum number of VLANs allowed in Cisco's standard VLAN range?

A) 128

B) 1005

C) 4094

D) 10000

Answer: B

Explanation: Cisco supports up to 1005 VLANs in standard range.

29.

Which VLAN is recommended to be changed from default for security?

A) VLAN 1

B) VLAN 10

C) VLAN 100

D) VLAN 4094

Answer: A

Explanation: VLAN 1 should be changed for security.

30.

Which feature prevents a switch port from forwarding BPDU frames?

- A) PortFast
- B) BPDU Guard
- C) VTP
- D) DTP

Answer: B

Explanation: BPDU Guard disables ports on BPDU receipt.

Hard (10)

31.

How does VLAN tagging affect the maximum transmission unit (MTU)?

- A) MTU decreases by 4 bytes
- B) MTU increases by 4 bytes
- C) MTU remains same
- D) MTU doubles

Answer: B

Explanation: VLAN tag adds 4 bytes to Ethernet frame, so MTU must accommodate.

32.

What is the function of the VLAN Trunk Protocol (VTP) pruning?

- A) Prevent VLAN pruning on trunks
- B) Dynamically block VLANs on trunks where unnecessary
- C) Increase VLAN numbers

D) Remove VLANs globally

Answer: B

Explanation: VTP pruning blocks unnecessary VLAN traffic.

33.

Which VLAN mode on Cisco switches does NOT participate in VTP but forwards VTP messages?

A) Server

B) Client

C) Transparent

D) Off

Answer: C

Explanation: Transparent mode forwards VTP messages but does not update VLANs.

34.

Which attack exploits VLAN tagging to gain access to unauthorized VLANs?

A) ARP poisoning

B) MAC flooding

C) VLAN hopping

D) DHCP starvation

Answer: C

Explanation: VLAN hopping exploits tagging.

35.

What is the main difference between ISL and 802.1Q VLAN tagging?

A) ISL is Cisco proprietary, 802.1Q is open standard

B) 802.1Q encapsulates whole Ethernet frame

C) ISL adds 4-byte header, 802.1Q adds 30 bytes

D) ISL does not support trunking

Answer: A

Explanation: ISL is Cisco proprietary; 802.1Q is IEEE standard.

36.

How do private VLANs increase security?

- A) By encrypting VLAN traffic
- B) By isolating ports within the same VLAN
- C) By tagging frames twice
- D) By increasing broadcast domains

Answer: B

Explanation: Private VLANs isolate ports on the same VLAN.

37.

Which Cisco command disables DTP on a switch port?

- A) switchport nonegotiate
- B) no switchport
- C) switchport mode trunk
- D) switchport access vlan 1

Answer: A

Explanation: switchport nonegotiate disables DTP negotiation.

38.

What is the effect of VLAN mismatch on a trunk link?

- A) No effect
- B) Increased throughput
- C) Traffic loss or misrouting for the mismatched VLAN
- D) Switch reboot

Answer: C

Explanation: Mismatched VLANs cause traffic loss on that VLAN.

39.

How does a Layer 3 switch differ in VLAN handling from a Layer 2 switch?

- A) Layer 3 switch can route between VLANs internally
- B) Layer 3 switch cannot handle VLAN tags
- C) Layer 3 switch only supports one VLAN
- D) Layer 3 switch does not support trunking

Answer: A

Explanation: Layer 3 switches route between VLANs.

40.

Which VLAN tagging field specifies priority?

- A) VLAN ID
- B) Priority Code Point (PCP)
- C) EtherType
- D) Subnet Mask

Answer: B

Explanation: PCP field specifies frame priority in 802.1Q tag.

41.

What is the primary purpose of a management VLAN?

- A) Carry voice traffic
- B) Isolate user traffic
- C) Manage network devices
- D) Carry routing protocols

Answer: C

Explanation: Management VLAN isolates management traffic.

42.

Which of these is NOT a valid VLAN range in Cisco switches?

- A) 1-1005
- B) 1006-4094
- C) 4095-5000

D) 1002-1005 (legacy)

Answer: C

Explanation: VLAN range above 4094 is invalid.

43.

What is the recommended best practice for securing VLAN trunk ports?

A) Enable all VLANs on trunks

B) Limit allowed VLANs on trunks to necessary ones only

C) Use VLAN 1 as native VLAN

D) Disable STP

Answer: B

Explanation: Limit VLANs to reduce attack surface.

44.

Which Cisco feature allows automatic VLAN membership assignment?

A) VLAN pooling

B) VMPS

C) VLAN pruning

D) VTP

Answer: B

Explanation: VLAN Membership Policy Server assigns VLANs dynamically.

45.

What does the VLAN ID 0 signify in an 802.1Q frame?

A) Default VLAN

B) Priority-tagged frame without VLAN info

C) Reserved for management

D) Native VLAN tag

Answer: B

Explanation: VLAN ID 0 means priority tagging without VLAN.

46.

How does the switch behave if MAC address table overflows due to MAC flooding?

- A) Drops frames
- B) Forwards all frames to all ports (flooding)
- C) Shuts down ports
- D) Filters MAC addresses

Answer: B

Explanation: Switch floods frames, enabling sniffing attacks.

47.

Which Cisco protocol can dynamically prune VLANs on trunks?

- A) VTP pruning
- B) DTP
- C) STP
- D) CDP

Answer: A

Explanation: VTP pruning restricts VLAN traffic.

48.

What is the main reason for changing the native VLAN from VLAN 1?

- A) Increase network speed
- B) Avoid VLAN hopping attacks
- C) Enable routing
- D) Enable VLAN pruning

Answer: B

Explanation: Changing native VLAN reduces VLAN hopping risks.

49.

Which VLAN encapsulation method inserts a tag inside the Ethernet frame?

- A) ISL
- B) 802.1Q
- C) MPLS
- D) PPP

Answer: B

Explanation: 802.1Q inserts tag inside Ethernet frame.

50.

What is the size in bytes of an 802.1Q VLAN tag?

- A) 2 bytes
- B) 4 bytes
- C) 6 bytes
- D) 8 bytes

Answer: B

Explanation: VLAN tag is 4 bytes.